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# Advanced Computer Architectures

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| Name | |
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| ID Number | |

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|---|---|
| **Problem 1 (20%)** | |
| **Problem 2 (15%)** | |
| **Problem 3 (15%)** | |
| **Question 1 (15%)** | |
| **Question 2 (20%)** | |
| **Question 3 (15%)** | |
| **Total (100%)** | |

**NOTE: students have 90' to complete the exam.**
**To pass the exam, a minimum score of 25% is requested by each "section" (section 1: problems, section 2: questions).**

## Problem 1

In this problem, you will port code to a simple **4-issue VLIW machine,** and schedule it to improve performance.
Details about the 4-issue VLIW machine with 4 fully pipelined functional units:

- Integer ALU with **1** cycle latency to next Integer/FP and **2** cycle latency to next Branch
- Memory Unit with **3** cycle latency
- Floating Point Unit with **3** cycle latency (each FPU can complete one add or one multiply per clock cycle)
- Branch completed with **1 cycle delay slot** (branch solved in ID stage)

**Assembly Code:**

```
Bb0: ld      $1, 24($R1)
     ld      $2, 8($R1)
     ld      $3, 4($R1)
     addd    $2, $2, $3
     addd    $1, $1, $2
     addi    $R1 $R1 4
     sd      $1, 16($fp)
     bnez $R1, BB0
```

**P1.A** Considering **one iteration** of the loop, **schedule** the assembly code for the 4-issue VLIW machine in the following table by using the **list-based scheduling,** but neither using software pipelining nor loop unrolling **nor modifying loop indexes.** Please do not need to write in NOPs (can leave blank).

**P1.B** Answer the following performance questions

**How long is the Critical Path?**

**What performance did you achieve in FP ops per cycle?**

## Answer 1.A

| | ALU 1 cc | MEM 2 cc | MEM 2 cc | FPU 5cc |
|---|---|---|---|---|
| C1 | | ld $1, 24($R1) | ld $2, 8($R1) | |
| C2 | addi $R1 $R1 4 | ld $3, 4($R1) | | |
| C3 | | | | |
| C4 | | | | addd $2, $2, $3 |
| C5 | | | | |
| C6 | | | | |
| C7 | | | | |
| C8 | | | | |
| C9 | | | | addd $1, $1, $2 |
| C10 | | | | |
| C11 | | | | |
| C12 | | | | |
| C13 | | | | |
| C14 | bnez $R1, BB0 | sd $1, 16($fp) | | |
| C15 | br delay slot | | | |

## Answer 1.B

**How long is the Critical Path?**

**What performance did you achieve in FP ops per cycle?**

## Problem 2

Please consider the program in the table be executed on a CPU with dynamic scheduling based on TOMASULO. Consider an architecture with:

- 3 RESERVATION STATIONS (RS1, RS2, RS3) + 2 LDU units (LDU1, LDU2) with latency 3 cycles
- 2 RESERVATION STATIONS (RS4, RS5) + 2 FPU (FPU1, FPU2) with latency 5 cycles

| | Instruction | ISSUE | Start EXE | WB |
|---|---|---|---|---|
| I1 | ld $1, 28($fp) | 1 | 2 | 5 |
| I2 | ld $2, 16($fp) | 2 | 3 | 6 |
| I3 | addd $1, $1, $2 | 3 | 7 | 12 |
| I4 | ld $2, 12($fp) | 4 | 6 | 9 |
| I5 | addd $2, $1, $2 | 5 | 13 | 18 |
| I6 | addd $1, $1, 1 | 13 | 14 | 19 |
| I7 | sd $1, 0($2) | 14 | 20 | 23 |

## Problem 3

Consider the following access pattern on a two-processor system with a direct-mapped, write-back cache with one cache block and a two cache block memory. Assume the **MESI protocol** is used, with **write-back** caches, **write-allocate,** and **write-invalidate** of other caches.

| Time | After Operation | Px cache block state | Py cache block state | Pz cache block state | Memory at Block 0 up to date? | Memory at Block 1 up to date? | Memory at Block 2 up to date? |
|---|---|---|---|---|---|---|---|
| 1 | Px read block 0 | Exclusive (0) | Invalid | Modified (2) | yes | yes | no |
| 2 | Py write block 1 | Exclusive (0) | Modified (1) | Modified (2) | yes | no | no |
| 3 | Px read block 1 | Shared (1) | Shared (1) | Modified (2) | yes | yes | no |
| 4 | Pz read block 1 | Shared (1) | Shared (1) | Shared (1) | yes | yes | yes |
| 5 | Pz write block 1 | Invalid | Invalid | Modified (1) | yes | no | yes |
| 6 | Px read block 2 | Exclusive (2) | Invalid | Modified (1) | yes | no | yes |